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Remagnetization Age and Mechanism of Cretaceous Sediments in Relation to Dyke Intrusion, Hainan Island: Tectonic Implications for South China and the Red River Fault

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Key Points:

- Chemical remagnetization of Cretaceous sedimentary rocks in Hainan is dated between 106.6 Ma and 104.6 Ma
- Authigenic growth of iron oxides and sulfides in the Cretaceous sediments occurred within 2 Myr of deposition
- The South China Block has remained fixed to Eurasia since the Cretaceous (105 Ma)

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Hainan Island lies near the Red River Fault, a prominent tectonic feature produced by the India-Asia collision. There, we carried out a geochronologic and paleomagnetic study on Cretaceous rocks in order to better understand the kinematic history of the region. U-Pb zircon dating of tuff intercalated in red bed sedimentary rocks yielded a concordant age of 106.6 ± 0.3 Ma; a mafic dyke intruding the red beds yielded a concordant age of 104.6 ± 0.7 Ma. Stepwise demagnetization experiments on 448 sedimentary rock samples and 191 dyke samples isolate solely normal polarities. Paleomagnetic directions of the dykes cluster in two distinct populations in geographic coordinates, indicating that dyke intrusion occurred in two pulses of limited duration (secular variation was not averaged) after tilting of the sediments. Baking of the sediments from the dykes only occurred near the contacts. Together with published data, the mean directions of 104 sites most tightly group at 58.3 ± 3.2% unfolding, indicative of a synfolding remagnetization, which can be constrained to have occurred within a 2 Myr period between sedimentation and dyke intrusion. We suggest that warm (50–100°C) fluid interaction during basin development led to new mineral growth spawning chemical remagnetization. The corresponding paleomagnetic pole at 81.5°N, 145.2°E (A_{95} = 2.4°) is indistinguishable from the coeval Eurasian reference pole, suggesting the South China Block has remained fixed to Eurasia since 105 Ma. A contour map of paleomagnetic rotations from 115 studies in the region shows that the Red River Fault roughly demarcates rotation magnitudes/signs, suggestive of a major tectonic boundary.

Plain Language Summary Highly stable magnetic remanences of hematite-bearing sedimentary rocks (red beds) form the basis of numerous paleomagnetic studies. However, an on-going scientific debate called the "red bed controversy" stems from the relative timing when the remanence was acquired (primary = detrital vs. secondary = chemical). Our article concerns a paleomagnetic and geochronologic study of Cretaceous red beds and cross cutting dykes from Hainan Island (South China). Radiometric-dating of tuff intercalated in the red beds and the crosscutting mafic dykes, combined with paleomagnetic analyses of 639 samples, allows us to precisely constrain the age of a pervasive synfolding remagnetization at ca. 105 Ma. Together with electron microscopy, we demonstrate the existence of a chemical remanent magnetization (CRM) that occurred within 2 Myr of the depositional age of the sediments. We also show that the dykes had limited thermal influence on the wall rocks. Importantly, the red beds that have CRMs do not suffer from inclination shallowing—the paleolatitude is very compatible with that predicted from the reference curve, suggesting the relative configuration between South China and Siberia has remained constant since 105 Ma.

1. Introduction

The India-Asia collision produced regional-scale, strike-slip faults in southeast Asia, which helped facilitate the extrusion of crustal blocks since the Eocene (Tapponnier et al., 1982). A primary tectonic feature facilitating extrusion is the >1,000 km long Red River Fault, which displaced Indochina southeast relative to South China (Leloup et al., 2001). According to the extrusion model, the Red River fault accommodated >700 km of sinistral motion (Leloup et al., 2001) and facilitated the opening of the South China Sea (Briais et al., 1993). This model predicts that Indochina rotated clockwise relative to South China, as corroborated by paleomagnetic data (Yang & Besse, 1993). A key constraint for the amount of motion on the Red River fault comes from distinct Mesozoic geochemical and structural features (Gan-Hang Zone, Shiwandashan Basin, etc.) that are truncated by the Cenozoic

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Red River Fault. Reconciling where the same features lie south of the fault led Gilder et al. (1996) to suggest that the fault displaced Indochina by more than 1,000 km relative to South China.

Extrusion accommodated by 1,000 km of motion on the ca. 315°N-trending Red River Fault would shift Indochina 700 km south of its pre-extrusion position (Tapponnier et al., 1982), which should be resolvable with paleomagnetism. A major obstacle with Cretaceous paleomagnetism in this part of the world is that most Cretaceous rocks are continental red beds, which are notoriously prone to inclination shallowing (Narumoto et al., 2006). The average inclination of Cretaceous and Paleogene strata from South China is 20 ± 5° lower than predicted from the Eurasian Apparent Polar Wander Path (Li et al., 2017). Taken at face value, this would suggest South China moved >1,000 km north with respect to Eurasia in the last 60 Ma, which seems highly improbable. Imprecise paleogeographic positioning of South China hinders comparisons with Sundaland, and hence inclination shallowing obscures paleomagnetic estimates of the motion along the Red River Fault.

Pre-cretaceous strata from South China often yield negative fold tests (Gao et al., 2019; Kent et al., 1987; Zhang et al., 2020). The remagnetizations have normal polarities with inclinations that are steeper than the Cretaceous sediments. The remagnetization is likely chemical in nature and chemical remanent magnetizations (CRMs) should not be prone to inclination shallowing. However, it is hard to constrain the timing of remagnetization, making these data difficult to interpret for tectonic studies.

Early Mesozoic paleomagnetic data from South China exhibit both clockwise and counterclockwise rotations of highly variable magnitude. Although rotation amounts in Cretaceous strata are much more subdued, streaking of the paleopoles implies South China experienced a certain degree of shear, which several workers attribute to far-field effects from the India-Asia collision (Gilder et al., 1993; Kawamura et al., 2013; Li et al., 1995; Liu & Morinaga, 1999). A major goal is to work out how the India-Asia collision deformed South China in space and time, which motivated our present study.

The center of Hainan Island lies ∼250 km north of the Red River Fault at the southernmost margin of the South China Block (Figure 1a). Cretaceous red beds from Hainan have been investigated by paleomagnetism (Fu, 2010; Fu et al., 2010; Li et al., 1995; Liu & Morinaga, 1999; Mo & Shi, 1987; Zhang & Tan, 2011), yet uncertainty persists whether the magnetizations are primary (Fu et al., 2010; Li et al., 1995) or secondary (Liu & Morinaga, 1999; Zhang & Tan, 2011), which bears on tectonic interpretations. For example, Mo and Shi (1987) analyzed eight samples with no field test. Li et al. (1995) studied 55 samples from six sites and interpreted the magnetization as primary, although the tightest grouping of the directions occurs at 71.5% unfolding. Liu and Morinaga (1999) studied 11 sites whose Fisher (1953) precision parameter (k) maximized at 25% unfolding. Combined with the data of Li et al. (1995), k maximized at 54% unfolding. Three samples in Liu and Morinaga (1999) were throughout to have reversed polarity, however, the radius of the 95% confidence ellipse of the three is >30°, which should not be considered reliable. Fu (2010) and Fu et al. (2010) studied 38 sites of Cretaceous red beds from Qionghai, Baisha, Ledong, and Wenchang (Figure 1a); k maximized at 100% unfolding thereby signifying a primary remanence, whereas seven sites from Dingan held synfolding magnetizations with k maximizing at 51% unfolding. Fu et al. (2010) also described a preliminary baked contact test on two dykes, where paleomagnetic directions in the samples farther from the dykes are distinctly different from the dykes. Zhang and Tan (2011) conducted a paleomagnetic and magnetic anisotropy study on 14 sites of Lumushan red beds in the same area as Liu and Morinaga (1999). They found k maximized at 77.0 ± 12.2% unfolding; however, our analysis of the same data (high temperature component, Table S3 in Supporting Information S1) shows k maximizes at 70.1 ± 13.0%, incompatible with a prefolding remanence. Distinct tectonic fabrics determined by anisotropy of magnetic susceptibility led Zhang and Tan (2011) to argue that the paleomagnetic data were overprinted.

Given the ambiguity surrounding Cretaceous paleomagnetic data from Hainan, and their importance for the tectonics of South China, we further investigated the problem to complement/understand existing results. Here we report new Cretaceous paleomagnetic data as well as radiometric dating of an intercalated tuff horizon within the sampled sediments and a mafic dyke that intruded the sediments to constrain when the remanent magnetization was acquired. We also show how baking from the dykes influences the magnetization of the sediments. We then compile the exhaustive paleomagnetic data from the island and place them in context of the tectonic evolution of South China.

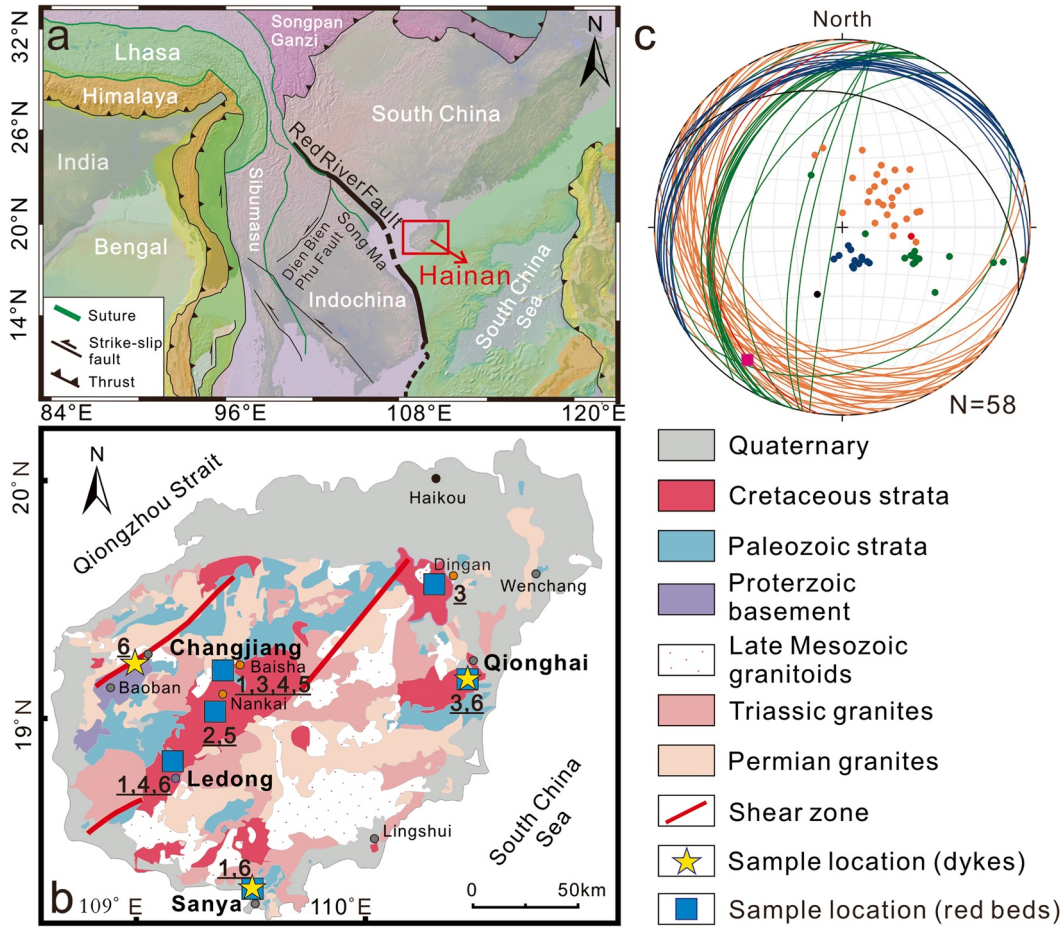


Figure 1. Geology of Hainan Island. (a) Simplified topographic and tectonic map showing the main tectonic units of Southeast Asia (after Leloup et al., [2001] and Li et al., [2017]). (b) Geological map of Hainan Island (after Wang et al., [1991] and Zhang et al., [2011]) showing faults/shear zones and the study localities (stars for dykes and squares for red beds). Numbers 1–6 correspond to paleomagnetic sampling locations (Table S3 in Supporting Information S1): (1), Li et al. (1995); (2), Liu and Morinaga, (1999); (3), Fu et al. (2010); (4), Fu (2010); (5), Zhang and Tan (2011); (6), this study. (c) Lower hemisphere stereonet plot of the bedding attitudes from Ledong, Qionghai, Baisha, Sanya, and Changjiang (data from this study, Li et al., [1995], Liu and Morinaga, [1999] and Zhang and Tan [2011]) and upper hemisphere projection of the poles to bedding defining a fold/tilt axis direction/plunge of 214.3°/11.3° (red square). The colors of the great circles and poles correspond to Ledong (green), Qionghai (blue), Baisha-Nankai (orange), Sanya (black), and Changjiang (red).

2. Geology and Sampling

Cretaceous extension produced the Southeast China Basin and Range Province (Gilder et al., 1991), whose NE-trending red bed basins were controlled by rifting (Shu et al., 2009); the Cretaceous basins in Hainan follow the same structural trend. NE-trending sinistral shear faults often bound the Cretaceous basins at Baisha, Ledong, Dingan, and Qionghai (Figure 1b; Metcalfe, 1994; Wang et al., 1991; Zhang et al., 2011). For example, the Gezhen fault near Changjiang in NW Hainan contains kinematic indicators such as S–C fabrics and mica fish (Zhang et al., 2011).

We carried out a geochronologic and paleomagnetic study of the Cretaceous Lumuwan Formation in the Qionghai, Sanya, Changjiang, and Ledong areas (Figure 1b). Unconformably overlying Paleozoic strata or Permian-Triassic granitoids, the Lumuwan Formation consists of fluvial-lacustrine, gray-purple/gray-green sandstone to conglomerate, with intercalated silty mudstone, tuff, shale and marl (Figure S1a-b, i-k in Supporting Information S1). A number of NE-trending diabase dykes (Figure S1c-h in Supporting Information S1) intruded the Lumuwan Formation along joints (Figure S1c-e in Supporting Information S1) or along strike slip faults as

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evidenced by horizontal slickensides in the Qionghai area (Figure S1f in Supporting Information S1); dykes intruded Mesozoic granite around Sanya and Changjiang (Figure S1g-h in Supporting Information S1).

We collected samples in the Lumuwan Formation for radiometric dating from a tuff in Ledong (18.8°N, 109.2°E) and a diabase dyke in Qionghai (19.1°N, 110.4°E). For paleomagnetism, at Ledong we drilled 17 sites of fluvio-lacustrine marls ($n = 86$) and sandstones ($n = 118$) in stratigraphic succession (~120 m; Figure S1a-b in Supporting Information S1), with each site spanning 2–5 m to average secular variation. At Qionghai (Figure S1c-f in Supporting Information S1), we drilled 137 samples in eight diabase dykes with thicknesses ranging from 15 to 400 cm; 2–4 profiles of 135 samples in the host sandstone were drilled at set distances away from four dykes to determine the influence of baking on the magnetization of the sediments. We also collected 109 sandstone samples from 11 sites that lie >10 m away from any observed dyke. Moreover, we sampled two dykes ($n = 30$) near Sanya (18.4°N, 109.4°E, Figure S1g in Supporting Information S1) and two dykes ($n = 24$) near Changjiang (19.3°N, 108.9°E, Figure S1h in Supporting Information S1) that were dated by Dilek and Tang (2021) as mid-Cretaceous (ca. 101 Ma). Samples were oriented by magnetic and Sun compasses (when possible) to correct core and bedding azimuths. The average of 38 and 100 Sun compass readings in 2017 and 2021 (respectively) yielded an average magnetic declination anomaly of $-0.7 \pm 2.2°$ and $-2.2 \pm 2.4°$ which agree with the predicted international geomagnetic reference field values of $-1.7°$ for 2017 and -2.0 for 2021.

3. Microscopy

To examine the textural relationships and diagenetic conditions of the magnetic minerals, polished thin sections from the dyke, marl, and sandstone samples were investigated with a petrographic microscope and a Zeiss SUPPA 55 field emission scanning electron microscope (SEM) at the China University of Geosciences (Beijing). Elemental concentrations were determined using energy dispersive spectrometry (EDS).

Thin section observations of sandstone reveal compositions mainly of lithic arkose and feldspathic litharenite (Figure S2a-b in Supporting Information S1). The sandstones are generally submature, with a <5% muddy (clay and silt) matrix and angular to subangular grains that are moderately sorted, sometimes cemented by calcite. Numerous large grains (>100 μm) of hematite exist in the particle-supported red sandstones (Figure S2a-b in Supporting Information S1). They are subangular to subrounded, dark red with red rims at high illumination; they have concavo-convex contacts with quartz and feldspar grains (Figure S2a-b in Supporting Information S1). Hematite pigment forms thin rims on grain surfaces (Figure S2a-b in Supporting Information S1), being responsible for the red color of the rocks. Some ferromagnesian silicates are partially or wholly altered to fine-grained hematite (Figure S2b in Supporting Information S1). Detrital grains such as quartz, feldspar, opaque minerals, and fossil fragments are present in the marl (Figure S2c in Supporting Information S1).

At the Ledong section, we collected gray-green rhyolitic tuff (sample Pm1-U-Pb-04) that is mainly composed of volcanic crystal fragments together with a minor amount of nonvolcanic detrital material, indicating deposition through the water column (Figure S2d-e in Supporting Information S1). Crystal fragments comprise about 55%, consisting mostly of quartz followed by feldspar and biotite. Most of the quartz and feldspar are angular to subangular. Biotite grains are flaky and oriented parallel to bedding. Detritus (~10%) is composed mainly of chert and mudstone. Yellowish brown cement is argillaceous, 30%–35% of the whole rock.

Diabase from Qionghai (Pm2-d-03) is dark green with an ophitic texture (Figure S2f in Supporting Information S1). Plagioclase (~70%) and clinopyroxene (~30%) are the main minerals; plagioclase forms small needles with weak sericitization, while clinopyroxene is granulous and chloritized. Ophitic textures are common with angular interstices between needle shaped plagioclase grains occupied by pyroxene or opaque minerals. In addition, some chloritized clinopyroxene phenocrysts (Figure S2f in Supporting Information S1) with an abnormal blue interference color are conspicuously larger than in the fine-grained groundmass and have a porphyritic texture.

SEM observations focused on opaque, iron-bearing minerals (Figures 2 and 3), with the phase interpretation based on the habit, the EDS analyses (Figures S3-S4 in Supporting Information S1) and the demagnetization characteristics (Figure 5). Opaque minerals in a 25-cm-thick dyke from Qionghai are (titano)magnetite, mostly <20 μm (Figures 2a–2f). A marl sample from Ledong contains widely distributed, 2–10 μm euhedral iron sulfides (Figure 3a). EDS analyses suggest that most is pyrite (Fe:S = 0.4–0.5), while some overgrowth rims (Figures S4g–h in Supporting Information S1) may be greigite or an oxidized phase with higher Fe:S (0.8–1.0).

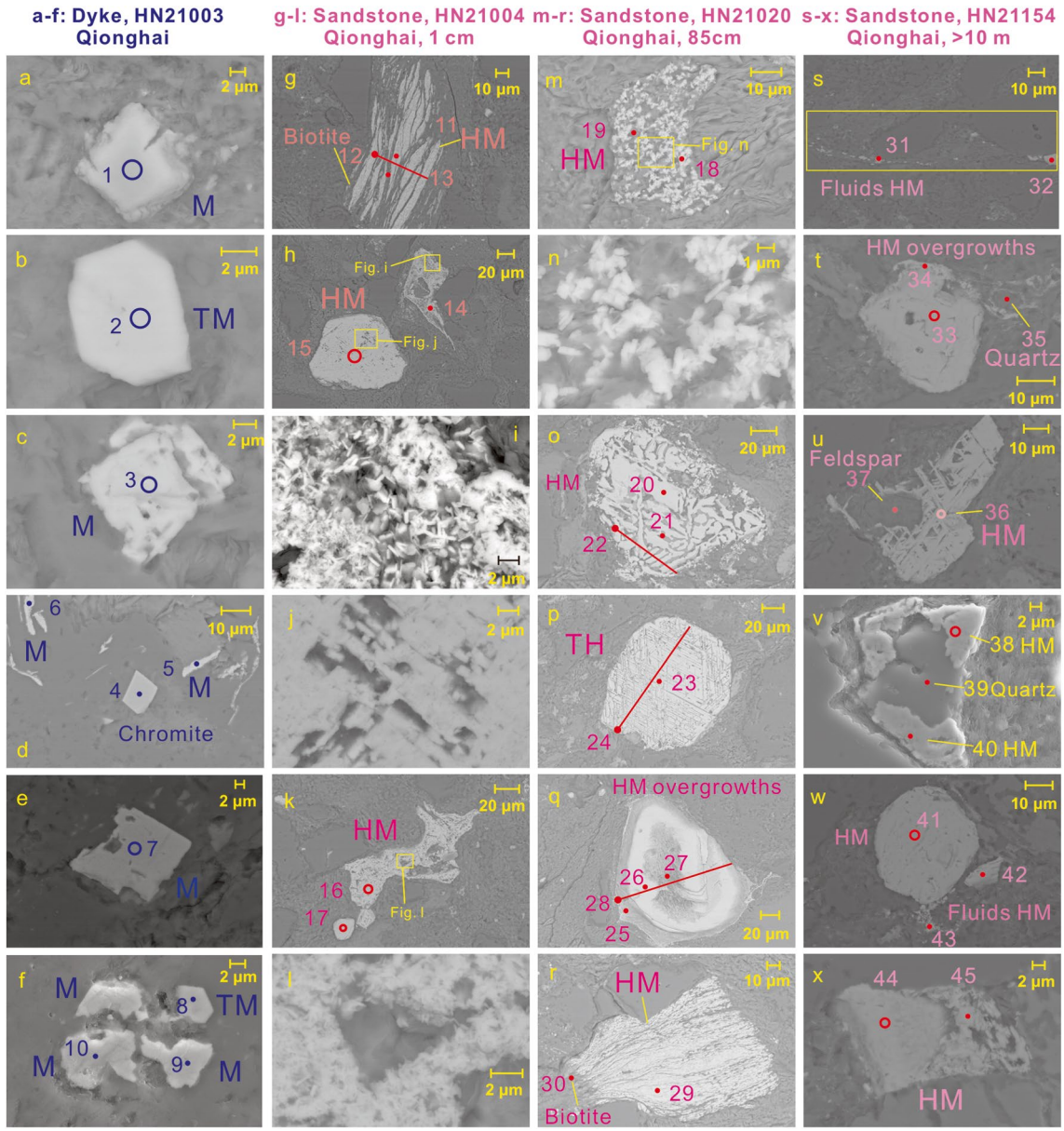


Figure 2. Scanning electron microscope (SEM) images of the 25 cm-thick dyke from Site 1A and red sandstone samples at distances from the dyke of 1 cm, 85 cm and >10 m in Qionghai. (a–f) SEM photomicrographs showing (titano)magnetite grains in a mafic dyke. (g–x) Authigenic hematite in red sandstones. Enlargement of areas (i, j, l, n) outlined in (h, k, m) showing oblate clusters of microcrystalline hematite. Silicates have been pervasively replaced by hematite in (g, m, r, u). A cluster of hematite encrusts the surface of a quartz or hematite overgrowths in (q, t, v, x). (s, w) Authigenic microcrystalline hematite in interstitial voids. The dots or lines with numbers indicate corresponding quantitative EDS spectrum analysis shown in Figure S3 in Supporting Information S1. HM: hematite, TH: titanohematite, M: magnetite, TM: titanomagnetite.

Framboidal pyrite aggregates are the dominant form and occur in voids in the matrix (Figures 3a and 3e). Clearly discernible are outlines of former framework grains (rutile in Figures 3b, c, e; titanomagnetite in Figure 3d) that have been dissolved in situ and are rimmed by authigenic pyrite. For the red sandstone samples from both Qionghai and Ledong, (titano)hematite is present as micro-to monocrystalline grains or as crystal aggregates (Figures 2g–2x; 3j–r). The hematite crystals vary in size from <1 to >100 μm; they are unequivocally authigenic at all studied distances from the 25 cm-thick dyke (1 cm, 85 cm, >10 m) in Qionghai, and in Ledong where no dykes are present. They form in secondary voids (Figures 2s and 2w) and replace detrital iron-bearing silicate minerals (Figures 2g, r; 3l). Platy microcrystalline hematite crystals typically range around 1–2 μm (Figures 2i, j, l, n; 3n, o, p)—they are present as interlocking networks (Figures 2j; 3j, n), as oblate clusters (Figures 2i and 2l),

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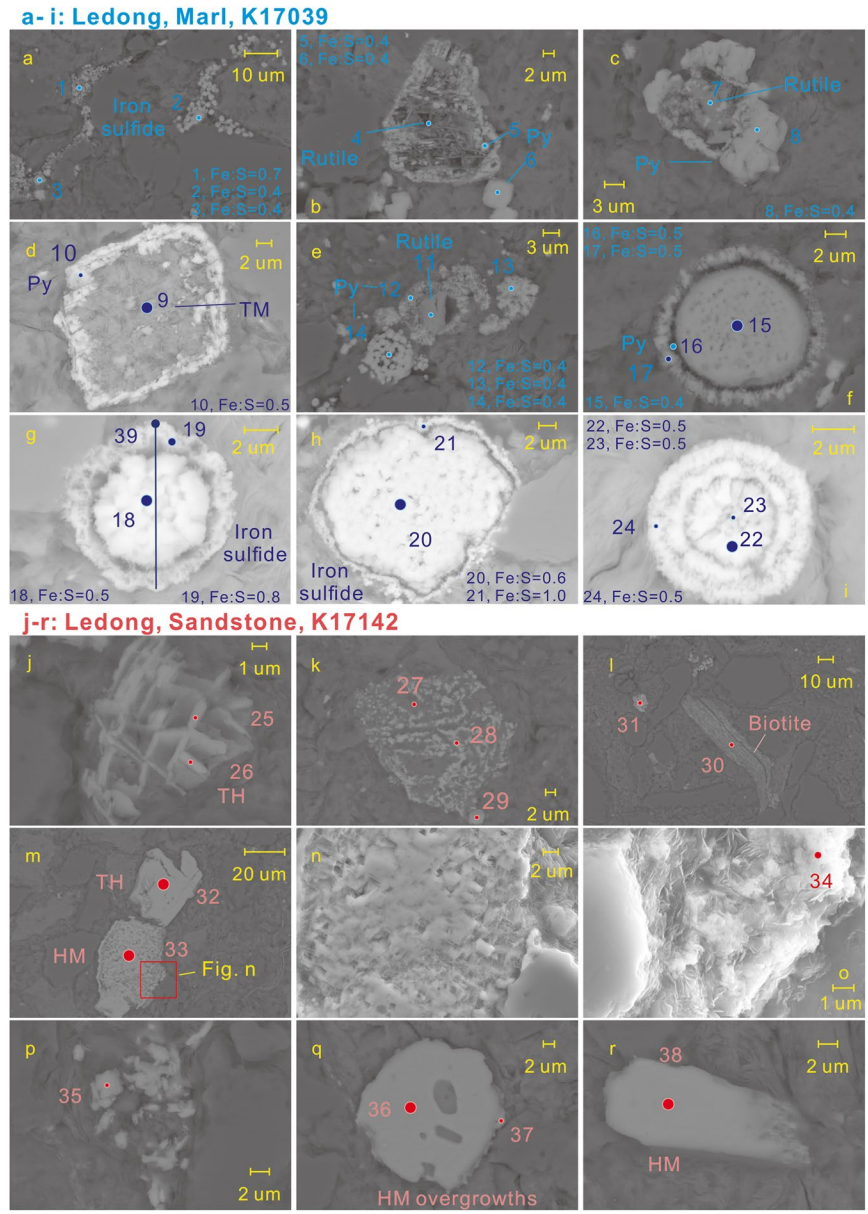


Figure 3. Field emission scanning electron microscope images of marl and red sandstone samples from Ledong. (a–i) Bright backscattered electron images are iron sulfide, including framboidal aggregates and overgrowths. Most of the iron sulfide is pyrite based on Fe:S = 0.4–0.5, while the overgrowths in (g–h) may be greigite or oxidized phases with higher Fe concentrations. (j–r) Authigenic (titano)hematite crystals distributed in a red sandstone sample similar to those in Qionghai. Enlarged view of area outlined in (n–o) shows aggregates of microcrystalline hematite. (k–l) Silicate grains pervasively replaced by hematite. (q) Crystalline hematite with overgrowths. Dots with numbers in the images indicate the corresponding quantitative EDS spectra shown in Figure S4 in Supporting Information S1. HM: hematite, TH: titanohematite, Py: pyrite, TM: titanomagnetite.

as drusy coatings on the surfaces of framework grains (Figures 2n; 3°, p), and as overgrowths on other hematite grains (Figures 2t, x; 3q). When associated with altered biotite, the microcrystalline hematite concentrates along cleavage lamellae (Figures 2g, r; 3l). Places where microcrystalline hematite grains are highly concentrated often cause the biotite flakes to splay apart (Figures 2g, r; 3l). Furthermore, microcrystalline hematite is superimposed on minerals such as feldspar and quartz (Figures 2u, v, t). Authigenic microcrystalline hematite also commonly exists as overgrowths on (titano)hematite grains (Figures 2t, q, w, x; 3q). Inasmuch as transportation would round off such protuberances, their presence suggests that the overgrowths grew in situ.

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The microscopic evidence indicates that iron sulfide and most of the hematite in the Lumuwan Formation sediments formed authigenically, after deposition of the sediments. The authigenic hematite most likely carries a CRM. Magnetite was only observed as large detrital grains; that sub-micron detrital or authigenic magnetic or greigite (if present) carry the remanence in some samples cannot be ruled out.

4. Geochronology

Zircon crystals were separated from the tuff and diabase samples using standard density and magnetic separation techniques. The zircon grains were handpicked and mounted in an epoxy resin disk, and then polished and coated with gold. Transmitted and reflected light photomicrographs as well as cathodoluminescence (CL) images indicated the zircon's external morphology and internal structure; these were used as a guide to select locations for in situ spot analyses. U-Pb isotopic analyses were performed with a Thermo Fisher's X-Series 2 laser ablation inductively coupled plasma mass spectrometer at the China University of Geosciences, Beijing using a Coherent GeoLasPro-193 nm laser. All data were acquired in single spot ablation mode with a spot size of 32 μm at 6 Hz; trace elements were calibrated with a National Institute of Standards and Technology (NIST, USA) standard SRM610, zircon 91500 was used as an external standard for U-Th-Pb isotopic ratios (Wiedenbeck et al., 2004). Either Plešovice or Mud Tank zircons were used as a monitoring standard (Sláma et al., 2008). Off-line selection and integration of background and analytical signals, time-drift correction and quantitative calibration for U-Pb dating were performed using ICPMS Data Cal (Ludwig, 2008). U-Pb analyses with high age discordance (>10% from the mean) were excluded.

Fifty-one zircon grains separated from the tuff were euhedral, without abrasive imprints, and exhibit ellipsoidal morphology or long prismatic outlines (Figure 4a). Mean grain sizes were ca. 75 μm (width) ×150 μm (length; Figure 4a). Th/U ratios range from 0.35 to 0.83 and cluster around 0.43–0.60 (Table S1 in Supporting Information S1). These zircons have positive Ce anomalies and small negative Eu anomalies (Figure 4c). CL images show that the majority of zircons have oscillatory compositional zoning and few core-mantle structures or overgrowth rims (Figure 4a), thereby indicating that the analyzed zircons were not inherited but grew in the primary melt (Belousova et al., 2002). The $^{206}Pb/^{238}U$ weighted mean eruption/deposition age from 51 spot analyses is 106.6 ± 0.3 Ma (Figures 4b and Table S1 in Supporting Information S1), which is compatible with the late-Early Cretaceous age based on ostracoda and charophyte fossils (Wang, et al., 1991), sporopollen assemblages (Huang & Long, 2008), and the youngest detrital zircon age (~120 Ma; Jiang et al., 2015) obtained from the Lumuwan Formation in the Ledong-Baisha area. The youngest detrital zircon ages in Lumuwan red beds are 105.7 ± 2 Ma in Baisha (19.4°N, 109.7°E) and 108.3 ± 2.8 Ma in Qionghai (19.1°N, 110.4°E; Chen et al., 2021), which are compatible with the tuff age of 106.6 ± 0.3 Ma in Ledong.

Most zircon crystals separated from the diabase fall in the size range of 40 μm (width) ×100 μm (length). CL images show that the majority of zircons have oscillatory compositional zoning (Figure 4d). No overgrowths, mineral or fluid inclusions, or metamictization was observed in the analyzed zircons, which indicate the absence of post-magmatic processes potentially disturbing a closed isotopic system. Th/U ratios vary from 0.40 to 1.39 (Table S1 in Supporting Information S1); positive Ce anomalies and negative Eu anomalies (Figure 4f) are indicative of a magmatic origin (Belousova et al., 2002). Nine analyzed spots on inherited zircons yielded Proterozoic to Late Triassic ages (228–1365 Ma). The ages of these inherited zircons are Meso-Proterozoic (1365 ± 13 Ma), Late Ordovician (459 ± 5 Ma), Silurian (423–432 Ma), and Late Triassic (228–233 Ma). The other nine magmatic zircons gave a $^{206}Pb/^{238}U$ weighted mean age of 104.6 ± 0.7 Ma (Figures 4f and Table S1 in Supporting Information S1), which we interpret as the best estimate for the crystallization age of the diabase. The Early Cretaceous age of the Qionghai dyke is compatible with U-Pb SHRIMP dating of 101 ± 4 Ma on the same dyke system (Dilek & Tang, 2021).

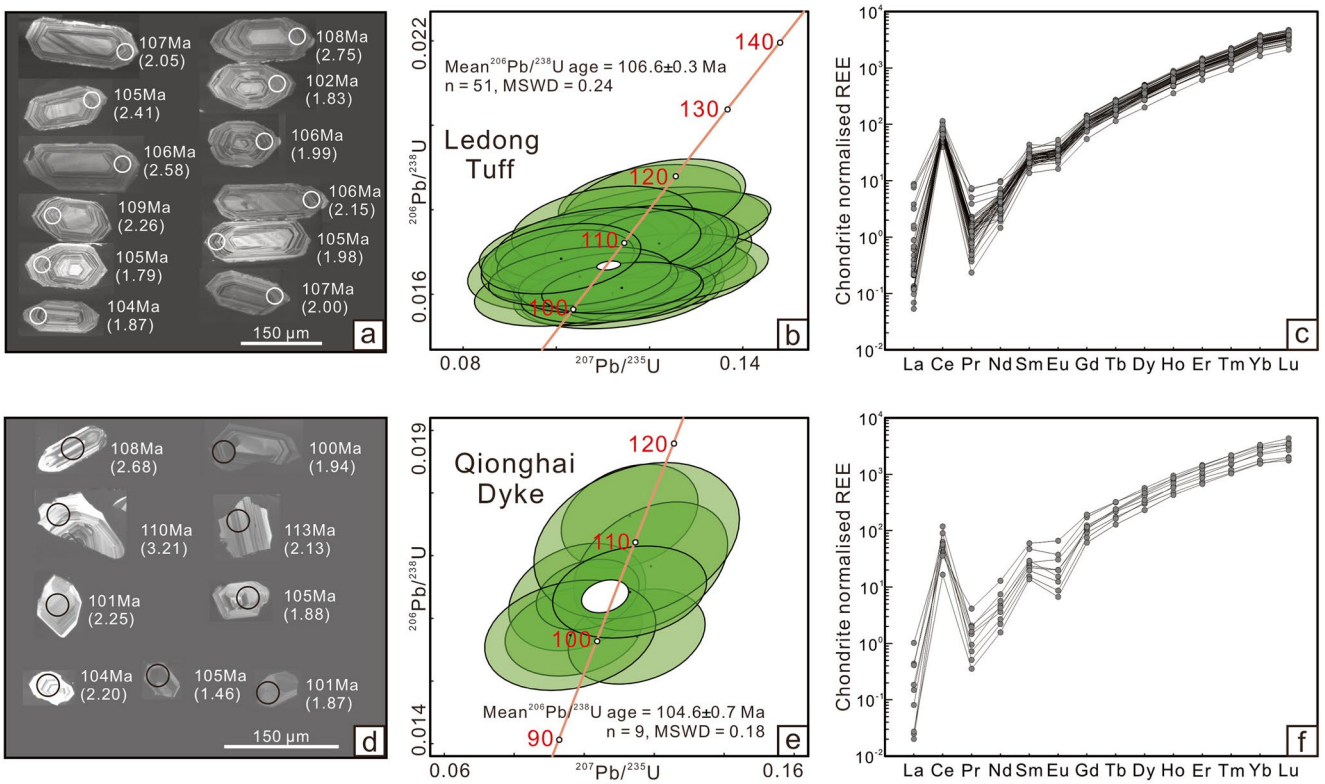


Figure 4. Representative cathodoluminescence (CL) images showing the internal structure of the analyzed zircon, U-Pb concordia diagrams and chondrite normalized REE patterns of Lumuwan Formation (a-c) tuff (Ledong) and (d-f) diabase (Qionghai). Mean concordia age = the maximum likelihood estimate of the age using the algorithm of Ludwig (1998) (1σ uncertainty); MSWD = the mean square weighted deviation.

5. Paleomagnetism

Demagnetization treatment and remanence measurements were carried out in a magnetically shielded room with residual fields less than 500 nT at the South China Sea Institute of Oceanography. Magnetic remanence was measured with a 2G Enterprises, three-axis cryogenic magnetometer using the software of Wack (2010). Thermal demagnetization was performed up to 570°C for the marl samples and 680–700°C for the sandstone samples in steps that were initially as large as 50°C but decreased to 10–20°C in and around the unblocking temperatures of magnetite and hematite. Sister samples of marl and sandstone were stepwise demagnetized using a time-decaying alternating field (AF) with peak amplitudes up to 150 mT. Software programs PMGSC (Enkin, 1994), PaleoMac (Cogné, 2003), www.paleomagnetism.org (Koymans et al., 2020), and PmagPy (Tauxe et al., 2016) were used for principal component analysis (Kirschvink, 1980), Fisher statistics (Fisher, 1953), and virtual geomagnetic pole (VGP) calculations (Coe et al., 1985).

The natural remanent magnetization (NRM) values of 171 dyke samples range from 2.0×10^{-3} to 6.4×10^{-1} A/m with a median of 1.1×10^{-1} A/m. Figures 5a–5d show the typical demagnetization behavior of the dykes; thermal and AF demagnetized sister specimens have indistinguishable directions. A high temperature component (HTC) was isolated >300°C with final unblocking temperatures of 500–580°C, indicating the presence of low titanium titanomagnetite as the main remanence carrier, consistent with the SEM observations (Figures 2a–2f; Table S2 in Supporting Information S1). All directions are north and down directed, indicative of magnetizations acquired during the Cretaceous normal superchron. The overall mean direction of 12 dyke sites from three locations is $D_g = 6.2°$, $I_g = 43.7°$ ($k_g = 143.0$, $\alpha_{95g} = 3.6°$) in geographic (g) coordinates and $D_s = 359.6°$, $I_s = 28.3°$ ($k_s = 74.5$, $\alpha_{95s} = 5.1°$) in stratigraphic (s) coordinates (Figures 6a-6b; Table S3 in Supporting Information S1; *D*, declination; I, inclination; k, the best estimate of the precision parameter [Fisher, 1953]; α_{95}, radius of a cone about the mean set at 95% confidence limits). The conversion to stratigraphic coordinates was based on the bedding attitudes of the sediments (Figure S1i-k in Supporting Information S1) cross cut by the dykes. The

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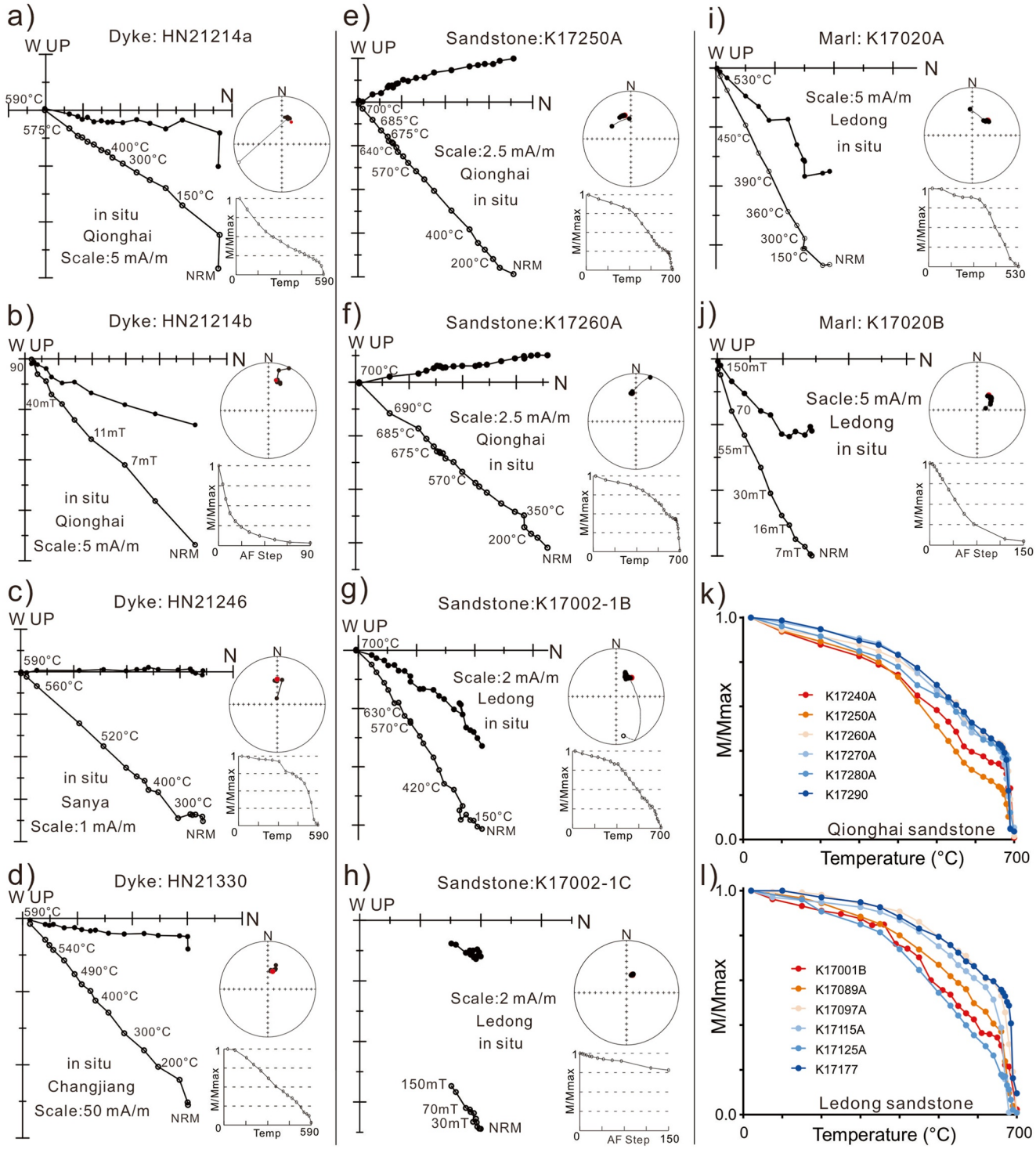


Figure 5. Stepwise demagnetization data for representative specimens from this study: vector-component diagrams, normalized magnetic moment versus temperature/alternating field and equal-area stereonet plots (in-situ coordinates). (a–d) Dykes from Qionghai, Sanya, and Changjiang; (e–f) red sandstones from Qionghai that lie >10 m from any observable dyke; (g–h) red sandstones from Ledong; (i–j) marls from Ledong. Solid (open) circles are projections in the horizontal (vertical) planes in the vector-component diagrams. Open (solid) circles indicate upper (lower) hemisphere in the stereonet plots. (k–l) Normalized moment decay plots for thermally magnetized red sandstones from Qionghai (samples >10 m from nearest dyke only) and Ledong.

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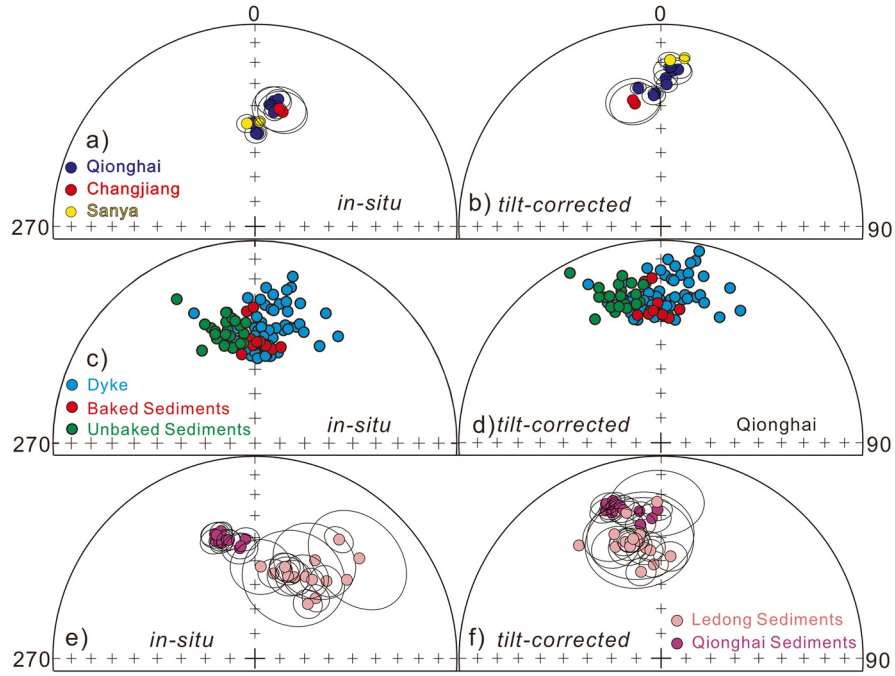


Figure 6. Equal-area projections of the high temperature component directions from this study in geographic (in situ) and stratigraphic (tilt corrected) coordinates. (a and b) Site mean directions from dykes in Qionghai (blue), Sanya (yellow), and Changjiang (red). (c and d) Individual sample directions separated as a function of distance from four dykes as described in the text. Blue are sample directions from the dykes, red are sample directions from sediments lying within 5 cm of the dyke and green are sample directions from sediments lying >20–80 cm from the dyke. Means (stars) and α_{95} shown for each population. (e and f) Site means from sediments collected >10 m from any visible dyke from Ledong and Qionghai.

McFadden (1990) Xi1 fold test is inconclusive; k maximizes at 30% unfolding (199.4) and an inclinations-only fold test is negative—k is 121.9 at 0% unfolding and 69.9 at 100% unfolding.

A closer look at the data reveals that the site mean directions of the dykes form two tightly clustered groups in geographic coordinates: one with a mean at $D_g = 359.9°$, $I_g = 49.2°$ ($k_g = 663$, $\alpha_{95} = 3.0°$) and another at $D_g = 10.1°$, $I_g = 39.6°$ ($k_g = 871$, $\alpha_{95} = 2.0°$; Figures 6a-6b). Independently, the fold test is negative for each cluster with k_{max} at 0% unfolding. The negative fold test and high k values lead us to conclude that the dykes intruded in two discrete pulses of short duration (e.g., secular variation was not averaged) after the sediments were tilted.

In order to determine the influence of dyke intrusion on the remanence acquisition mechanism of the host sediments, we drilled the neighboring red beds as a function of distance away from the dykes along multiple profiles (Figures 7a-7b). Stepwise demagnetization of the sediments displays one or two magnetization components (Figures 7c–7f). Moment decay plots of the thermal demagnetization data indicate an enhancement of the magnetite fraction (more pronounced decay by 580°C) relative to the hematite fraction approaching the dykes (Figures 7g–7i). The paleomagnetic directions of the red beds closest (<5 cm) to the dykes agree well with the mean direction of the corresponding dyke; going farther away, the directions in the sediments deviate from those of the dykes (Figures 6c-6d). In general, thermal demagnetization of samples collected several 10s of cm from a dyke distinguishes directions at the lower unblocking temperatures that lie closer to the dyke directions in geographic coordinates while directions at higher unblocking temperatures trend more northwesterly with shallower inclinations (Figures 7c–7f). Sediments closest to the dykes have higher NRM intensities compared to samples collected farther from the dyke (Figures 7j-7l). The rate of decrease in NRM with distance is proportional to the dyke thickness: high NRMs persist 10–20 cm from the 25 cm-thick dyke, while high NRMs are found 25–80 cm from the 400 cm-thick dyke (Figures 7j-7l; Table S4 in Supporting Information S1). Normalizing the x-axis (distance from dyke) on Figures 7j-7l by the dyke thickness shows a fairly universal pattern in moment decay, albeit with scatter, with 1/2 decrease in moment by 10% of the relative distance and 2/3 by 20%–50%.

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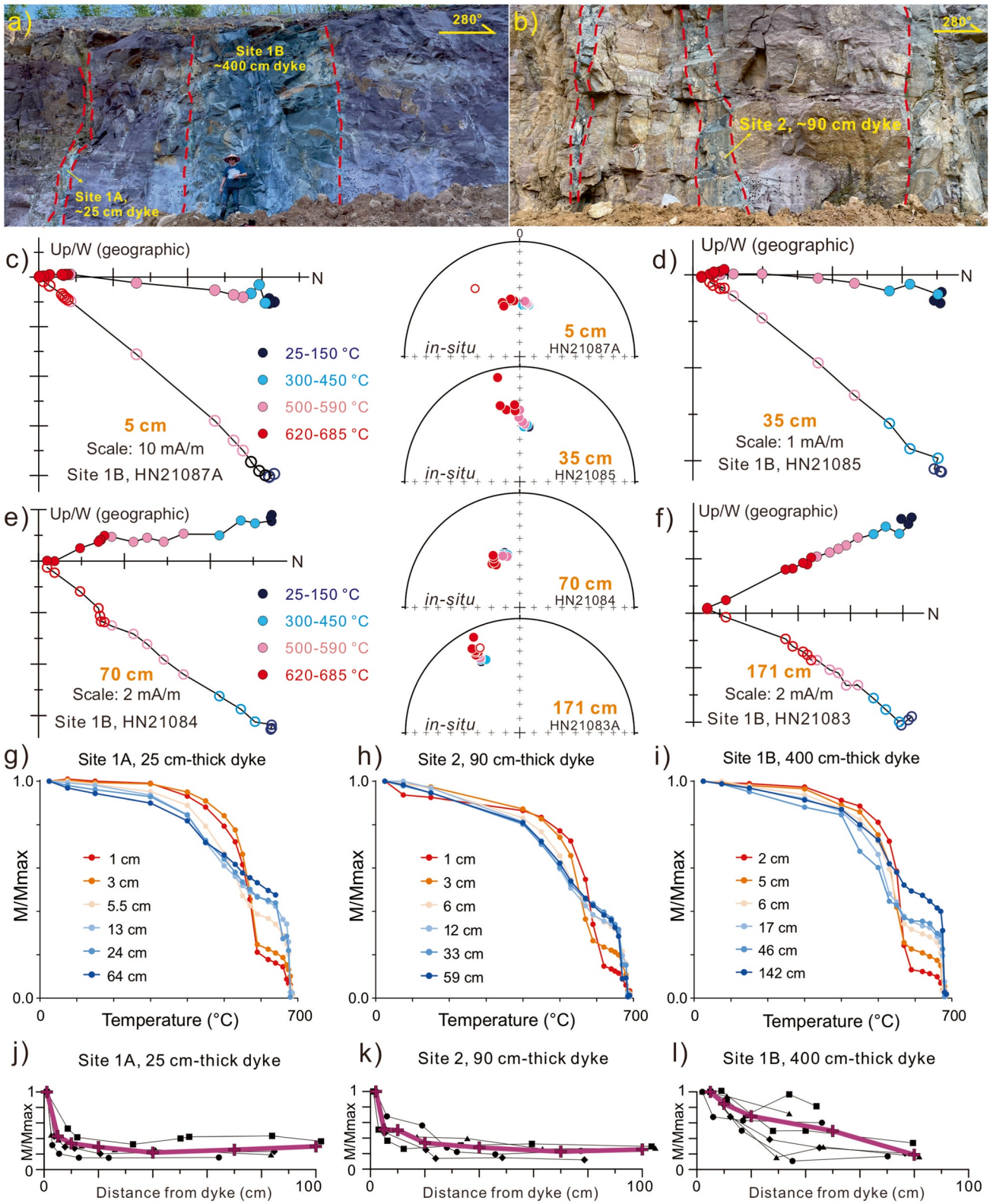


Figure 7.

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High NRM intensities in the samples closest to the dykes likely reflect thermal overprinting (i.e., a thermal remanent magnetization, TRM). A slight difference in direction below and above the Curie temperature of magnetite in samples farther from the contact could be attributed to temperatures that were high enough to impart TRMs in magnetite (~580°C) but not high enough to reset hematite (~680°C). Therefore, the baked contact test is positive based on the decrease in NRM intensity away from the contact with the dykes (Figure 7) and given that the dyke and neighboring red beds carry similar magnetization directions, whereas the unbaked red beds have significantly different directions than the dykes (Figures 6c-6d). It is possible that heat and fluids from dyking reduced hematite to magnetite in the nearby sediments. However, magnetite occurs naturally in the red beds well away from the dykes (Figure 5k), so it is more likely that thermal overprinting (TRM) of the existing magnetite produced much higher magnetizations than that generated by the original detrital remanence (DRM) or possibly a chemical remanence. Knowing that DRM intensities are much less than TRM intensities, the baking might have accentuated the intensity of the pre-existing magnetite, making it more visible in the moment decay plots.

Stepwise demagnetization of 109 sandstone samples from Qionghai that lie >10 m away from any visible dyke have NRM intensities ranging from 1.3×10^{-2} to 3.7×10^{-2} A/m with a median at 2.2×10^{-2} A/m. Thermal demagnetization isolates one dominant magnetization component of normal polarity that decays linearly to the origin from low temperatures up until complete demagnetization around 680°C (Figures 5e-5f). Normalized moment decay plots show a slight drop around 580°C and a sharp drop around 680°C (Figure 5k), suggesting that the magnetic remanence is carried by magnetite and hematite and that both carry similar magnetization directions.

Stepwise demagnetization of the Ledong sediments isolates one or two linear magnetization components, both of solely normal polarity. NRM intensities for 118 sandstone (34 marl) samples range from 1.3×10^{-3} to 4.2×10^{-2} A/m (2.9×10^{-4} to 4.2×10^{-2} A/m) with a median at 1.9×10^{-2} A/m (7.0×10^{-4} A/m). Thermal demagnetization of the sandstone samples isolates two types of unblocking spectra: (I) a wide range of unblocking from 400 to 680°C or (II) a sharp drop around 680°C (Figure 5l). Both decay by a least-squares fit straight to the origin (Figure 5g). In these sandstone samples, AF demagnetization removes only 25% of the NRM (Figure 5h), suggesting the dominance of a high coercivity phase, and together with the 680°C peak unblocking temperature, suggests the magnetic remanence is carried mostly by hematite.

A magnetization component of the 34 marl samples was identified as a linear segment in the demagnetization trajectory within unblocking temperatures from 300 to 530°C (Figure 5i) or 16–150 mT (Figure 5j). Roughly 75% of the NRM was removed by 90 mT (Figure 5j), suggesting the presence of a low-coercivity mineral like magnetite. Thermal and AF demagnetization trajectories and directions for sister marl samples are in good agreement (Figures 5i-5j). The mean direction of the 34 marl samples ($D_g = 40.2°$, $I_g = 53.9°$, $k_g = 48.9$, $\alpha_{95g} = 3.6°$; $D_s = 353.4°$, $I_s = 50.2°$, $k_s = 42.4$, $\alpha_{95s} = 3.8°$) is distinct at 95% confidence limits from the 118 red sandstones samples ($D_g = 25.3°$, $I_g = 49.0°$, $k_g = 26.9$, $\alpha_{95g} = 2.6°$; $D_s = 345.9°$, $I_s = 38.7°$, $k_s = 34.2$, $\alpha_{95s} = 2.3°$) at Ledong in either coordinate system. However, the distribution of the marl directions completely overlaps a portion of the directions from the red beds, so the difference in overall means of the two facies could arise from under sampling the marls.

Taken together, 261 HTCs isolated from the Ledong and Qionghai sediments (Table S2 in Supporting Information S1) pass the non-parametric fold test of Tauxe and Watson (1994) with k_{max} at 100% unfolding; however, an inclinations-only fold test is negative—k is a maximum at 0% unfolding (32.6) and a minimum at 100% unfolding (22.3). The 261 HTCs comprise 17 sites at Ledong and 11 sites at Qionghai, with each site having a single bedding attitude in order to compute a parametric fold test, noting that one site has two samples and two have three (Table S3 in Supporting Information S1). Applying the parametric fold test of Watson and Enkin (1993) on the 28 site mean directions (1,000 simulations, 5° uncertainty on bedding) yields an optimum at 88% unfolding;

Figure 7. Magnetic response of neighboring sediments in relation to dyke intrusion. (a and b) Field photos of sites 1A (25 cm thick dyke), 1B (400 cm thick dyke), and 2 (90 cm thick dyke) together with Lumuwan sediments. (c–f) Vector-component diagrams and equal-area stereonets of sediment samples from site 1B at distances of 5, 35, 70, and 171 cm from the 400 cm thick dyke. Colored symbols corresponds to demagnetization temperature steps as indicated in the vector-component diagrams (c and e). Solid (open) circles are projections in the horizontal (vertical) planes in the vector-component diagrams. Equal-area stereonets correspond to each of the samples as indicated using the same color scheme to indicate the temperature range. (g–i) Normalized moment decay plots of select thermally demagnetized samples as a function of distance from the nearest dyke as indicated by color. (j–l) NRM intensity of the samples collected along transects extending perpendicular from three dykes: individual profiles as thin black lines, median values as thick red lines. The NRM magnetization is normalized by the NRM of the sample with the maximum (highest) magnetization along the particular transect, which is often the sample closest to the dyke. The thick magenta line is the average value of the thin black lines from the individual transects.

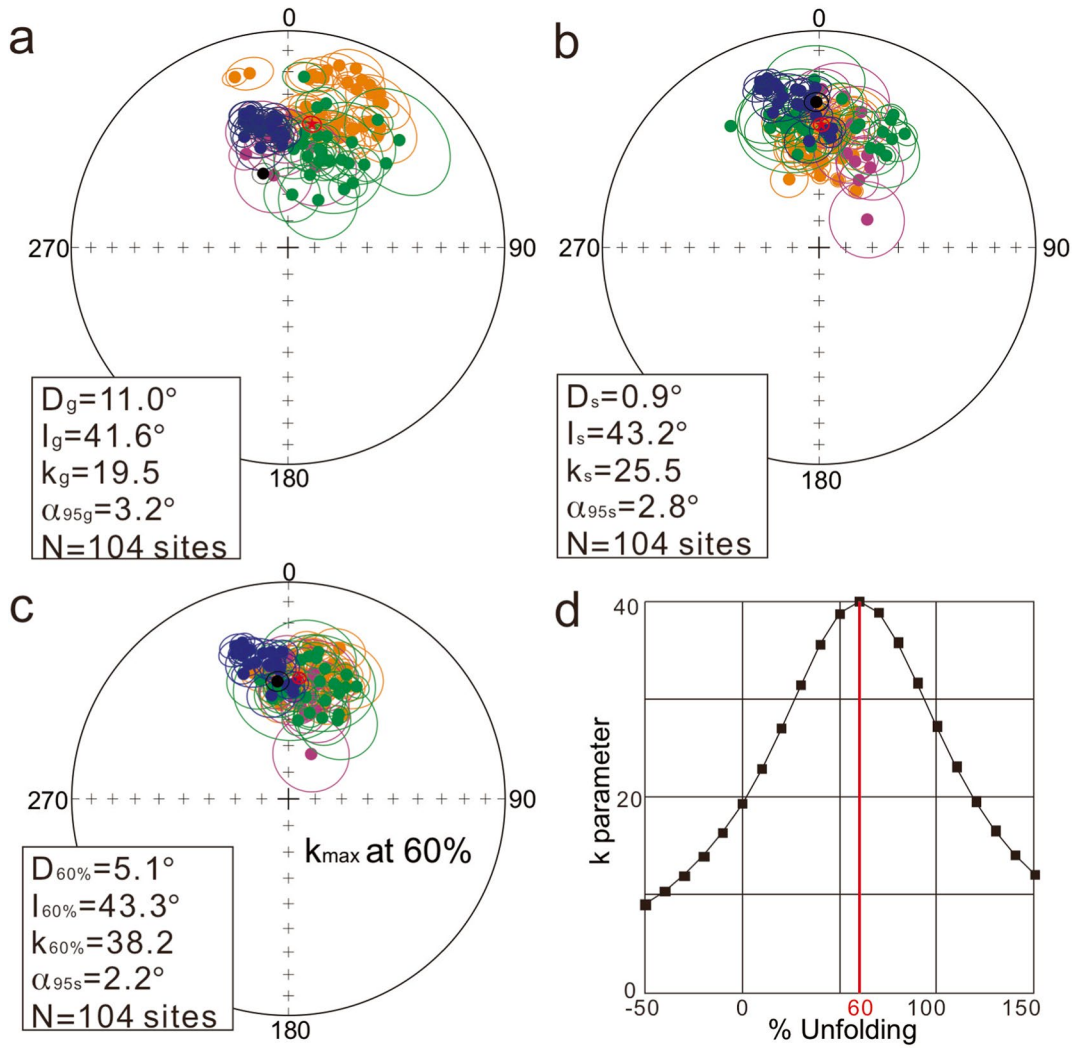


Figure 8. Equal-area projections of 104 site-mean directions for the Cretaceous Lumuwan Formation (Table S3 in Supporting Information S1) shown at (a) 0%, (b) 100%, and (c) 60% unfolding. Site mean colors and uncertainty ellipses correspond to Ledong (green), Qionghai (blue), Baisha (orange), Sanya (black), and Dingan (purple). (d) *k* versus percentage (%) of unfolding from the parametric fold test of Watson and Enkin (1993).

however, the parametric fold test on the inclination-only data (Enkin & Watson, 1996) yields an optimum at 7% unfolding.

We combined our 28 sites together with 76 other site-mean paleomagnetic data for the Lumuwan Formation from Hainan, together making an exceptionally large number of 104 sites (Table S3 in Supporting Information S1; Fu, 2010; Fu et al., 2010; Li et al., 1995; Liu & Morinaga, 1999; Zhang & Tan, 2011; this study). The Watson and Enkin (1993) fold test applied to these 104 sites yields an optimum degree of untilting at 58.3 ± 3.2% when running 1000 iterations with 5° uncertainty on bedding attitudes; the optimum degree of untilting from inclination-only data is 56.5 ± 5.0% (Enkin & Watson, 1996). Together, the results strongly argue for a synfolding magnetization at ~60% unfolding: $D_{60\%} = 5.1°$, $I_{60\%} = 43.3°$ ($k_{60\%} = 38.2$, $\alpha_{95} = 2.2°$; Figure 8), consistent with the original finding of Liu and Morinaga (1999). Using a two-step unfolding procedure that accounts for the slightly plunging fold axis (214.3°/11.3°) leads to a virtually identical result with a synfolding magnetization at 60%; in fact, Fisher statistics are even less favorable for the 100% unfolding case with *k* at 22.2 as opposed to 25.5 with the one-step procedure. Given the scatter of the bedding attitudes, we base further discussion on one-step unfolding for each site. Eliminating potential outliers from the data set; for example, after omitting all sites with

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$N \leq 3$ ($N = 98$ sites), yields no discernible difference—k_{max} still occurs at 60% unfolding. We see no logical way to divide the sites into subclasses to perform further testing.

6. Discussion

6.1. Age and Origin of the Synfolding Remagnetization in Hainan Island

Since the initial finding by Kent et al. (1987), remagnetized pre-Cretaceous strata with solely normal polarity, NNE declinations and moderate inclinations (40–60°) are commonly found throughout South China; however, they predominantly come from rocks that fail the fold test (e.g., Gao et al., 2019; Zhang et al., 2020). The age of remagnetization remains ill-defined. It is logical to assume the overprint component was acquired during the Cretaceous normal superchron based on the ubiquity of normal polarities and that the paleomagnetic poles roughly coincide with the Cretaceous reference pole from South China. For the Lumuwan Formation in Hainan, the remanence was clearly acquired during folding/tilting and all directions are solely of normal polarity. The overall mean direction at 0%, 60% or 100% unfolding (0%: $D = 11.0°$, $I = 41.6°$; 60%: $D = 5.1°$, $I = 43.3°$; 100%: $D = 0.9°$, $I = 43.2°$; Figure 8) are distinctly different from the present-day field ($D = 358.4°$, $I = 26.5°$) or geocentric axial dipole ($D = 0°$, $I = 34.6°$) directions, suggesting the synfolding overprint was not acquired during the Brunhes normal chron. The synfolding magnetization result means that the remanence is chemical and/or thermal in origin, possibly superimposed with a detrital remanent magnetization (DRM).

According to the baked contact test, thermal overprinting occurred only in samples lying in close proximity to the dykes, ruling out a regional thermal overprint. Moreover, because the remanence of the dykes was acquired after deformation and the sediments during deformation, the unbaked sedimentary rocks carry a remanence that predates dyke intrusion. Together with the authigenic hematite observed in the SEM images (Figures 2 and 3), we conclude the remagnetization is partly or wholly chemical in nature (CRM) and the CRM was acquired within 2 Myr of deposition (after the depositional age of 106.6 ± 0.3 Ma and before dyke intrusion at 104.6 ± 0.7 Ma). That the dykes contain primary, euhedral titanomagnetite with no hematite alteration further suggests the chemical remagnetization affected the sediments prior to dyke intrusion since the dykes contain insignificant amounts of hematite. Moreover, the titanomagnetite in the dykes reflects crystallization from primary melts with no/limited subsequent alteration. It is likely that the sediments contain overlapping DRM and CRM components. This might be reflected in the inclination data of the sedimentary sites from Ledong and Qionghai (Figures 6e-6f) where the sites have similar declinations but a spread in inclination. DRM-bearing sediments could be expected to have shallower inclinations than CRM-bearing sediments because the mechanisms leading to flattened inclinations in DRMs (mechanical interactions, pore space reduction, etc.) are largely absent with CRM.

In Hainan, the overall mean direction of the dykes in geographic coordinates ($D_g = 6.8°$, $I_g = 44.5°$, $k_g = 115.8$, $\alpha_{95} = 4.3°$) is indistinguishable at 95% confidence limits from the 60% synfolding direction of red beds ($D_{60\%} = 5.1°$, $I_{60\%} = 43.3°$, $k_{60\%} = 38.2$, $\alpha_{95} = 2.2°$), and the ages of the diabase dyke (104.6 ± 0.7 Ma) and tuff (106.6 ± 0.3 Ma) are close in time. The diabase dykes in Qionghai strike NE (Figure 1c), parallel to the NE-trending faults that facilitated basin formation. Moreover, Cretaceous NE-trending red bed basins in South China (e.g., Jianghan and Henyang) are commonly bound by NE-trending syn-sedimentary faults (Shu et al., 2009), which suggests basin development occurred simultaneously with fault formation that slightly predated dyke intrusion. No growth strata have been documented in the Hainan basins so far.

Magnetite and/or greigite in marl, and hematite with variable proportions of magnetite in sandstone, carry the syn-folding CRM magnetizations in Hainan. Several chemical overprint mechanisms have been proposed (e.g., Larson et al., 1982; Till & Nowaczyk, 2018; Walker et al., 1981). Concerning hematite, some studies found that large detrital hematite grains in red beds carry primary DRMs, whereas fine-grained pigmentary hematite carries a CRM (Tauxe, et al., 1980). Paleomagnetic, petrographic and geochemical studies have shown that faults can act as conduits for fluids that facilitate CRM (Elmore et al., 2006; Font et al., 2011). In North America, continent-scale remagnetization is associated with the end-stage of Pangaea formation during the Kiaman reversed superchron which caused the precipitation of various iron oxide phases depending on the facies and local chemical environment (Evans & Elmore, 2006; McCabe & Elmore, 1989).

South China is different in that the Cretaceous was a time of extension and basin formation, leading to the so-called Southeast China Basin and Range (SECBAR) province (Gilder et al., 1991). Zircon-based fission-track thermochronology data near NE-trending faults facilitating the opening of the Cretaceous extensional basins

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(e.g., Jianghan, Nanxiong, Boyang, Henyang, etc.) reveal a prominent, rapid cooling event at ~125–95 Ma (Wang et al., 2020), for example, during the Cretaceous normal superchron. We suspect that rapid unroofing of the horsts (cooling) during extension transferred heat to the nearby basins, thereby leading to warm (50–100°C) fluid circulation in the basins that facilitated partial to complete remagnetization over a protracted time period. Dykes in the Hainan basins likely signify that the geotherm was higher in Hainan, thereby further enhancing secondary growth. As the remagnetization occurred close to the time of deposition, it seems that the nature of the basin fluids played a key role in dictating the conditions necessary to convert ferrous (Fe^{2+}) to ferric (Fe^{3+}) iron to produce hematite, or the opposite to produce magnetite when carbon is present, as in the marls.

6.2. Paleogeography of South China and the Red River Fault Region

To better understand the deformation of the Red River Fault Region in response to the India-Asia collision, we compiled Cretaceous paleomagnetic results from recent compilations (Cung & Geissman, 2013; Jeong & Yu, 2019; Li et al., 2017), which presumably stem from primary magnetizations (Tables S5 and S6 in Supporting Information S1). Rotations and paleolatitude differences were calculated with respect to the synthetic apparent polar wander path (APWP) for Europe, which here we consider as Eurasia (Torsvik et al., 2012). For Hainan, the average paleomagnetic pole of the 104 sites at 60% unfolding is Plat = 81.5°, Plon = 145.2°, A_{95} = 2.4°, with a paleolatitude of 25.2 ± 1.7°N. With respect to the 100 Ma reference pole, Hainan is neither significantly rotated (−2.7 ± 5.2°) nor significantly displaced in latitude (flattening factor, $F = 2.0 \pm 3.8°$); hence, the paleomagnetic data argue that Hainan has remained fixed to Eurasia since the Cretaceous. We performed the same analyses for the 114 other studies listed in Table S6 in Supporting Information S1 using the reference pole that matches closest in time with the mean age of the rocks used in each study. Figure 9a shows a contour map of the rotations, while Figure 9b plots the paleolatitude differences (F) of the studies from South China.

Figure 9a contains several remarkable features. First, sites exhibiting counterclockwise rotations lie mostly northeast of the Red River Fault. Indeed, the limit between clockwise and counterclockwise rotations (0° contour) roughly parallels the Red River Fault. We interpret the clockwise/counterclockwise rotations in the region west/east of the Red River Fault as a result of left-lateral strike-slip activity at ~35–17 Ma during the southward extrusion of Indochina (Leloup et al., 1995; Li et al., 2017; Tapponnier et al., 1982). Likewise, counterclockwise rotations in Guangxi (study #s 5–7, 42) were attributed to left-lateral strike slip motion on a fault that runs parallel to the Red River Fault. The counterclockwise rotation around study #25 in Jiangxi province may be related to the Zhenghe-Dafu left-lateral fault (Li et al., 2014). Counterclockwise rotations farther north in Sichuan Province (study #s 23, 43, 45, 51, 52) can also be linked to deformation related to the India-Asia collision (Enkin et al., 1991). An unexpected outcome from the map is that most of central South China displays insignificant rotations (0° contour), whereas most of Sichuan province (eastern South China, i.e., #16–19) and the coastal provinces (western South China) possess clockwise rotations.

Hainan's Cretaceous (105 Ma) paleolatitude of 25.2 ± 1.7°N lies 6° north of its present latitude (19°N) and 12.4° ± 5.4° farther north than expected from the South China Block. However, Hainan's paleolatitude is indistinguishable with that predicted by the 100 Ma Eurasian reference pole (25.4 ± 3.3°N; Torsvik et al., 2012; Figure 9b). Most Cretaceous paleolatitudes from the South China Block are 10–20° lower than predicted by the Eurasian APWP (Figure 9b). These paleomagnetic data are likely based largely on DRMs obtained from red beds affected by inclination shallowing.

In contrast, the Hainan result is derived mostly from a CRM, which should be mostly immune to inclination shallowing, thereby explaining why the paleolatitude of the Hainan red beds agrees with the dykes as well as the expected paleolatitude from the synthetic Eurasian pole (#1, Figure 9b). Paleolatitudes derived from DRMs that were corrected for inclination shallowing (#24 and #25, Figure 9b) or stem from volcanic rocks (#3 and #14, Figure 9b) whose thermal remanences are likewise immune from shallowing artifacts also agree within 95% confidence limits of that predicted by the reference Cretaceous Eurasian APWP. If universally true, then DRM and CRM in red beds from South China can be discriminated by their paleolatitudes. Observed paleolatitudes that exhibit significant differences likely lack CRMs, and when not, then the magnetizations are likely CRM-based. Because CRMs are created within the first few Myr of deposition, the age of the reference pole used in the calculation will likely not bias the interpretations of the rotational component derived from Cretaceous red beds. Hence, we find that the South China Block experienced no significant latitudinal shift with respect to stable

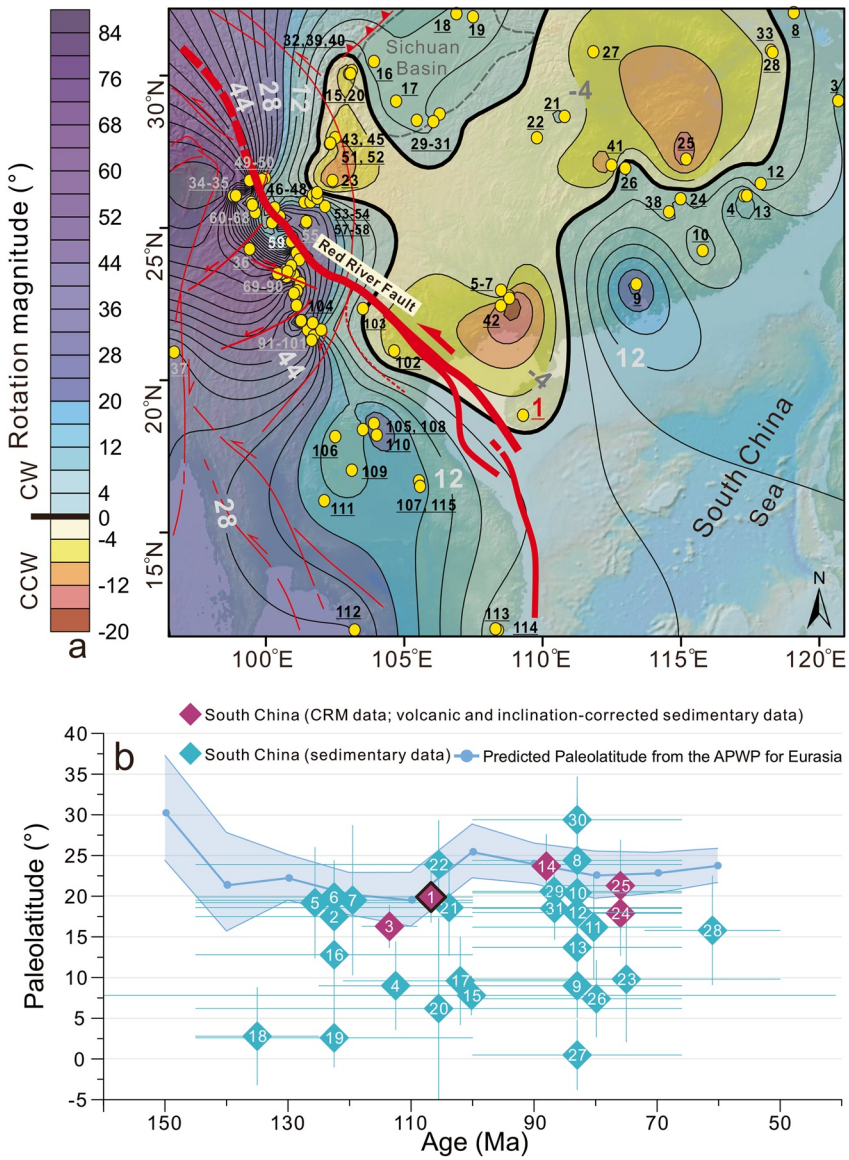


Figure 9. (a) Tectonic map with major faults of the SE Asian Blocks showing contours of rotation magnitude as indicated in the color scale (faults from Leloup et al., 2001; Li et al., 2017). Positive rotations are clockwise, negative rotations counterclockwise. Yellow circles denote paleomagnetic sampling sites (data in Table S6 in Supporting Information S1). The dashed gray line denotes the Sichuan basin. (b) Observed and expected Cretaceous paleolatitudes for South China. Numbers correspond to the paleomagnetic studies in Table S5 in Supporting Information S1. Purple diamonds represent studies based on sedimentary rocks whose remanences stem from a CRM (#1), DRMs corrected for inclination shallowing (#24 and #25) or TRMs from volcanic rocks (#3 and #14).

Eurasia since the Cretaceous. The Red River Fault represents a major lithospheric boundary between the South China and Indochina blocks as delineated by a marked difference in rotation sense and magnitude.

7. Conclusions

1. Non-marine, rift-related basins in Hainan Island formed in the late Early Cretaceous. U-Pb dating of a tuff intercalated in the Lumuwan Formation yielded 106.6 ± 0.3 Ma. Mafic dykes intruding the Lumuwan Formation were dated at 104.6 ± 0.7 Ma
2. Site mean directions from 12 dykes form two very tightly grouped populations in in-situ coordinates, each that fail the fold test. We interpret this to represent two brief pulses of intrusion after tilting of the host sediments

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3. 2 Myr or less passed between deposition and tilting of the red beds
4. Dyke intrusion had limited thermal/fluid influence on the wall rock, depending on the dyke thickness; NRM intensity and unblocking temperature spectra can be used to identify baked samples with TRMs
5. The *k* parameter of 104 site-mean directions of Cretaceous red beds from Hainan Island maximizes at 60% unfolding signifying a synfolding remagnetization. Scanning electron microscopy reveals authigenic hematite growth in the red beds, indicating the magnetic remanence is chemical in nature. It is probable that some of the samples in the Lumuwan Formanion possess original DRMs or a mix of DRM + CRM. Combined with the folding age of the sediments, we constrain the timing of the acquisition of the chemical remanence to within 2 Myr of original deposition. We think that remagnetization of the sediments occurred where uplifted horst blocks supplied heat to the nearby basins in a trans-tensional tectonic environment. The geothermal gradient was likely higher in Hainan than in other SECBAR basins as evinced by dyke intrusion
6. A contour plot of rotation magnitudes in southeast Asia reveal a highly differential rotation pattern along the Red River Fault, consistent with a major tectonic boundary. When considering only paleomagnetic data sets immune to inclination shallowing (volcanic rocks or CRM) or sediments that have been corrected for inclination shallowing, insignificant latitudinal motion is observed between the South China Block and stable Eurasia. Paleolatitude data arguing for significant N–S translation are derived from sedimentary rocks afflicted with inclination shallowing. A better assessment of the amount of N–S motion along the Red River Fault requires paleolatitude data from Sundaland that are unaffected by inclination shallowing

Data Availability Statement

Data sets for this research are available in the MagIC database (earthref.org/MagIC/19227). Competing interests: The authors declare no competing financial interests.

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